Electricity (Set B) — MCQ Worksheet

Science • Chapter 11 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Coulomb is the unit of:

(A) Charge
(B) Current
(C) Resistance
(D) Power

Q2. Charge on a single electron is approximately:

(A) -1.6×10^{-19} C
(B) 1.6×10^{-10} C
(C) 1 C
(D) 9×10^{-31} C

Q3. Ohm's law: V = IR is valid when:

(A) Physical conditions (especially temperature) remain constant
(B) Voltage is zero
(C) Current is zero
(D) Power is zero

Q4. Resistance of a conductor depends inversely on:

(A) Cross-section area
(B) Length
(C) Temperature
(D) Density

Q5. Resistance of a wire of length L and area A with resistivity ρ:

(A) $R = \rho L/A$
(B) $R = \rho A/L$
(C) $R = \rho LA$
(D) $R = \rho/(LA)$

Q6. If two resistors 6Ω and 12Ω are in series, equivalent R is:

(A) 18 Ω
(B) 4 Ω
(C) 2 Ω
(D) 8 Ω

Q7. Two resistors 6Ω and 12Ω in parallel give equivalent R:

(A) 4 Ω
(B) 18 Ω
(C) 2 Ω
(D) 8 Ω

Q8. Heat produced in a resistor is given by:

(A) $H = I^2Rt$
(B) $H = V^2/R$
(C) Both A and V^2t/R
(D) I × R only

Q9. Power in a device of resistance R carrying I is:

(A) $P = I^2R$
(B) P = V/I
(C) P = I/R
(D) P = V × t

Q10. Bulb of 100 W on 220 V draws current:

(A) ≈ 0.45 A
(B) ≈ 2 A
(C) ≈ 5 A
(D) ≈ 22 A

Q11. Fuse is connected in:

(A) Series with the appliance in live wire
(B) Parallel
(C) Anywhere
(D) Outside circuit

Q12. Earth wire is for:

(A) Safety, conducting leakage current to earth
(B) Increasing voltage
(C) Reducing voltage
(D) Storing electricity

Q13. Filament of bulb is made of tungsten because:

(A) High melting point and high resistance
(B) Cheap
(C) Light weight
(D) Cold

Q14. Increasing temperature usually:

(A) Increases resistance of metallic conductor
(B) Decreases
(C) No change
(D) Doubles

Q15. Domestic wiring in India operates at:

(A) 220 V AC, 50 Hz
(B) 110 V DC
(C) 5 V DC
(D) 240 V DC

Q16. Three resistors 2Ω, 3Ω and 6Ω in parallel give equivalent R:

(A) 1 Ω
(B) 5.5 Ω
(C) 11 Ω
(D) 2 Ω

Q17. A wire of resistance R is stretched to double its length. New resistance is:

(A) 4R
(B) 2R
(C) R/2
(D) R/4

Q18. A 60 W bulb is used 5 hours a day for 30 days. Energy used in kWh:

(A) 9 kWh
(B) 0.3 kWh
(C) 60 kWh
(D) 150 kWh

Q19. If a 100 W bulb costs 5 paise per hour, cost of running it 10 hours daily for 30 days at same rate:

(A) Rs 15
(B) Rs 5
(C) Rs 30
(D) Rs 50

Q20. Why is the heating element of an electric iron made of nichrome?

(A) High resistivity, high melting point, doesn't oxidise readily
(B) Cheap
(C) Light
(D) Conducts coolness

Q21. If voltage across a heater is doubled, power dissipated:

(A) Becomes 4 times ($P = V^2/R$)
(B) Doubles
(C) Halves
(D) Stays same

Q22. Resistance of a copper wire of length 1 m and cross section 0.01 cm^2 with $\rho = 1.6 \times 10^{-8}$ Ωm is:

(A) 0.016 Ω
(B) 1.6 Ω
(C) 16 Ω
(D) 160 Ω

Q23. Why is a parallel combination preferred for household appliances?

(A) Each gets full voltage and independent operation; if one fails, others work
(B) Saves wires
(C) Cheaper
(D) Increases voltage

Q24. If current of 2 A flows through a resistor of 5Ω for 1 minute, heat produced is:

(A) 1200 J
(B) 600 J
(C) 20 J
(D) 300 J

Q25. An MCB (Miniature Circuit Breaker) acts as:

(A) A reusable fuse, breaks circuit on overcurrent
(B) Voltage transformer
(C) Battery
(D) Capacitor

— End of Worksheet —